

P4 — GIS + AIS + Attribution

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

ROLE OBJECTIVE

Own AIS trajectory reasoning, GIS correlation and the explainable candidate-prioritisation features.

HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

PHASE 1 — DATA & PREPROCESSING

Work package 1: AIS data schema + cleaning

Define and validate MMSI/IMO, timestamp, latitude, longitude, SOG, COG, heading and vessel metadata; remove duplicates/out-of-range points; preserve source timestamps and provenance; document region/source limitations.

Deliverable: cleaned AIS schema + validation rules.

Dependencies / acceptance: Depends on selected AIS source. Acceptance: invalid observations are flagged/rejected and provenance is retained.

PHASE 2 — SPILL DETECTION

Work package 2: Geospatial detection handling

Transform segmentation geometry into geographic coordinates; validate coordinate reference systems; create spatial footprints for spill and source regions; prepare spatial queries for vessel intersections.

Deliverable: georeferenced spill/source geometries and spatial query functions.

Dependencies / acceptance: Depends on P1 segmentation and P2/PostGIS schema. Acceptance: geometry overlays correctly on the map and spatial queries return expected test cases.

PHASE 3 — SOURCE RECONSTRUCTION

Work package 3: GIS spatial preparation

Create spatial indexes, buffers/corridors and temporal-spatial query helpers; prepare basemap/reference layers; define distance calculations and coordinate conventions for evidence features.

Deliverable: reusable GIS utility layer.

Dependencies / acceptance: Depends on project CRS and P2 database. Acceptance: distance/intersection calculations pass controlled test cases.

PHASE 4 — AIS CORRELATION

Work package 4: Trajectory reconstruction + filtering

Order observations by vessel/time; detect gaps and discontinuities; interpolate only when justified; calculate distance to source corridor, time difference, heading/track compatibility and continuity; preserve AIS-quality flags.

Deliverable: reconstructed vessel trajectories + evidence features.

Dependencies / acceptance: Depends on P5 source corridor/time window. Acceptance: every candidate has traceable spatial, temporal and trajectory values.

PHASE 5 — EVIDENCE FUSION + UI

Work package 5: Explainable candidate features

Implement spatial proximity, temporal overlap, trajectory compatibility and AIS reception-quality features; define normalization and missing-data behaviour; provide feature-level explanations instead of a single opaque score.

Deliverable: candidate feature vector + explanation payload.

Dependencies / acceptance: Depends on P4 trajectory and P5 uncertainty. Acceptance: a reviewer can see why each feature increased/decreased investigative relevance.

PHASE 6 — VALIDATION & DEPLOYMENT

Work package 6: Attribution scenario testing

Create controlled synthetic vessel/source scenarios with known ground truth; run the candidate prioritisation process; calculate Top-1, Top-3 and MRR where appropriate; document cases where AIS gaps or ambiguity prevent a reliable ranking.

Deliverable: attribution evaluation dataset + results + failure analysis.

Dependencies / acceptance: Depends on P6 test design and P2 fusion service. Acceptance: ground truth is known by construction and results are reproducible.

CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.